

Dr Nisal Udawatta

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PROFESSIONAL SUMMARY

Experienced Computer Science lecturer with a strong academic and industry background, holding a PhD in Computer Science from Victoria University of Wellington. Skilled in interactive lecture delivery, course material preparation, curriculum development, and student mentoring, with teaching experience at the University of Kelaniya, Victoria University of Wellington, and Yoobee College of Creative Innovation. Experienced in delivering both in-person and online classes via MS Teams and Zoom to cohorts of over 100 learners across multiple campuses. Research expertise spans Human-Computer Interaction (HCI), User Experience (UX), and Extended Reality (XR) applications, with broader teaching strength across software development, databases, networking, and cyber security. Industry experience includes a Human-Machine Interface (HMI) internship at Trimble New Zealand, with hands-on proficiency in Agile/Scrum methodologies and technologies such as Unity, C#, web development, GitHub, UML, Figma, and AI.

EXPERIENCE

Lecturer – Technology | Yoobee College of Creative Innovation

Christchurch Campus • November 2025 – Present

Programme: Master of Business Informatics (MBI)

Courses: Business Strategy and Direction, Information Systems Analysis and Design, Database Management Systems

- Delivering Master's-level courses across the Christchurch and Auckland campuses, teaching cohorts of over 100 learners in-person and online via MS Teams and Zoom
- Preparing and refining course materials aligned to learning outcomes
- Mentoring students and serving as class-in-charge
- Supervising Master's-level capstone and applied research projects
- Contributing to curriculum development and moderation
- Maintaining an active research profile through ongoing publications
- Building industry communication and partnerships

PhD Candidate | Victoria University of Wellington

March 2021 – June 2025 • Supervisors: Dr Craig Anslow and Dr Brian Robinson

Research: Extended Reality to improve cardiac electrophysiology procedures

Domain: Human-Computer Interaction / Digital Health

Technologies: Unity, C#, Extended Reality (XR), Virtual Reality (VR), Mixed Reality (MR), Augmented Reality (AR), motion capture gloves, Git

- Conducting a thorough literature review and analysis to identify research gaps and opportunities
- Interviewing electrophysiologists in New Zealand (end users) to identify existing challenges
- Analysing interview data to identify end-user requirements
- Collaborating with end users to design the XR application, ensuring it meets their requirements
- Developing the XR application in Unity, applying 3D modelling, C# programming, and UI design
- Conducting evaluation studies to assess the effectiveness and usability of the XR application
- Writing a thesis documenting the research process, methodology, findings, and contributions to the field of electrophysiology

Tutor | Victoria University of Wellington

June 2023 – June 2025

Course Units: SWEN303 – User Experience Engineering; SWEN422 – Human-Computer Interaction

Technologies: UI/UX design principles and evaluation methods (e.g., Cognitive Walkthrough, Heuristic Evaluation)

- Marking essays, assignments, and examinations against established rubrics, providing constructive written feedback to students
- Advising students on academic and related matters through scheduled helpdesks

Human Machine Interface Research Intern | Trimble Navigation New Zealand Ltd

November 2024 – February 2025 • Managers: Grant Odgers and Darrin Hurd

Technologies: Unity, C#, Virtual Reality (VR), Miro, Git

- **Research and development:** Researched novel ways to improve Human-Machine Interface (HMI) and Human-Computer Interaction (HCI) for excavator machines

- **Prototype design:** Designed UI and developed software prototypes in Unity for Android; contributed concepts for a smart wearable in-cab device, down-selected the best ideas, built initial prototypes in a simulator, and conducted in-cab testing on a construction site
- **User feedback integration:** Iterated and refined prototypes rapidly based on user feedback
- **Usability evaluation:** Designed and conducted usability studies
- **Team collaboration:** Collaborated with team members to deliver project outcomes

Lecturer | University of Kelaniya

March 2018 – May 2021 • Domain: Computer Science

Technologies: C, C#, Unity, HTML, CSS, JS, PHP, Laravel, MS SQL, MySQL, Machine Learning, Cyber Security, Networking, Git, LMS (Moodle), Visual Studio, VS Code, VMware, Hyper-V

- Preparing and delivering lectures, tutorials, seminars, and laboratory sessions
- Courses taught: Introduction to Computers, Structured Programming (C) in Linux OS, Computer Systems Organisation, Data Communication and Networks, Computer Architecture and Design, Computer and Network Laboratory, Cyber Security Laboratory
- Preparing and marking essays, assignments, and examinations; invigilating exams
- Advising students on academic and related matters
- Attending departmental and faculty meetings, conferences, and seminars
- Supervising work programmes of honours students and tutorial staff
- Supervising final-year software development and research projects (B.Sc. (Hons) in Computer Science; Bachelor of Information and Communication Technology)
- Serving on faculty and other committees as Web Master and LMS (Moodle) Manager
- Managing the department website as webmaster (Joomla-based)
- Research and publications – [Google Scholar profile](#)

Trainee Software Engineer (Internship) | Bell Solutions (Pvt) Ltd

February 2016 – July 2016

Project: Insurance Claims Management System

Technologies: HTML, CSS, C#, ASP.NET with MVC architecture and Entity Framework, MS SQL, Tortoise SVN, Visual Studio

- Analysing client needs
- Designing and developing a web application for a Sri Lankan insurance broker to manage claims
- Testing, debugging, and resolving defects

EDUCATION

Ph.D. in Computer Science | Victoria University of Wellington, New Zealand • 2021 – 2025

B.Sc. (Special) in Computer Science | First Class Honours • University of Kelaniya, Sri Lanka • 2013 – 2017

ACHIEVEMENTS

- Third Place, International XR Workshop 2023 – Group project *XR for Palliative Care (EternalXR)*, organised by Victoria University of Wellington, University of Auckland, and University of Canterbury ([workshop site](#))
- IEEE VR Doctoral Consortium, 2022 – IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW)
- ACM ISS Doctoral Consortium, 2022 – Companion Proceedings of the 2022 Conference on Interactive Surfaces and Spaces
- Academic Excellence – Dean's List, 2013/2014 and 2014/2015, Faculty of Science, University of Kelaniya

VOLUNTEERING

- VR showcase volunteer at Victoria University of Wellington outreach event
- Student Volunteer, 2022 Conference on Interactive Surfaces and Spaces (ISS 2022 Companion), 20–23 November 2022, Wellington, New Zealand ([programme committee](#))

TECHNICAL SKILLS

Virtual Reality and Mixed Reality Development: Unity, C#, Microsoft Mixed Reality Toolkit, SteamVR, XR Interaction Toolkit, Oculus Integration, StretchSense motion capture gloves, Oculus Quest 2, Oculus Quest Pro, HTC VIVE Pro, Vuforia, OpenAI

Unity Development: VR – CardiacEPX, EternalXR; Games – Treasure Hunt Escape, Lava Scape

Web Development: ASP.NET (MVC), HTML, CSS, JavaScript, PHP, Laravel, PHPStorm, Visual Studio, VS Code

Databases: MS SQL, MySQL, Oracle

Other Programming Languages and Tools: C, C++ with OpenGL, VB.NET, Java, Python, PL/SQL, version control (GitHub, GitLab, TortoiseSVN), MATLAB

Teaching Delivery: MS Teams, Zoom, Moodle (LMS), Joomla, in-person and online delivery to large cohorts

PERSONAL SKILLS

- Research and development skills
- Analytical, communication, and interpersonal skills
- Self-management and team management
- Collaborative team player with a positive approach
- Eager to learn

References available upon request.